

TEST-01-MASTER-TEST-PLAN

TEST-01: MASTER TEST PLAN

Carrington Storm Motors — Project AEGIS

Comprehensive Verification & Validation Framework for All CSM Products

Document Status: Governing Document — All Test Protocols Derived From This Plan

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Classification: CONFIDENTIAL — Accredited Investor Eyes Only

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TESTING PHILOSOPHY

The CSM testing philosophy rests on four pillars:

- NIST-Traceable:** Every measurement is traceable to a National Institute of Standards and Technology (NIST) primary standard through an unbroken chain of calibrations. When CSM claims 92 dB shielding effectiveness at 45 µm MXene thickness, that claim is anchored to a NIST-calibrated field probe, a NIST-traceable network analyzer, and a NIST-traceable profilometer.
 - Peer-Reviewable:** All test data, methodologies, and analysis scripts are published in full (with proprietary manufacturing details redacted) to enable independent replication. CSM does not make claims that cannot be verified by a third-party laboratory following the published protocol.
 - Open-Data:** De-identified test results are made available to academic researchers, insurance actuaries, and regulatory bodies through the CSM Digital Twin platform’s data-sharing API. This commitment to transparency distinguishes CSM from defense contractors who classify comparable test data.
 - Carrington-Calibrated:** Every EMI/GIC test is calibrated to the measured parameters of the 1859 Carrington Event, scaled for modern infrastructure: surface electric fields of 1,000–5,000 mV/km, magnetic field change rates of 2,000 nT/min, and ground-induced current densities of 10–50 A/m².
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TEST CATEGORIES — FOUR-TIER HIERARCHY

Tier 1: Materials Testing (TEST-02)

Scope: Individual materials — MXene coatings, BFRP composites, ceramic dielectrics, geopolymer concrete, piezoelectric elements, aerogels, MR fluids, STF formulations

Standards: IEEE 299, MIL-STD-188-125, ASTM D149, ASTM D257, ASTM D4935, ASTM D3039, IEC 60243

Laboratory: CSM El Segundo QC Lab (TEM cell, universal test machine, environmental chambers) or accredited third-party lab

Sample Rate: Every production batch for critical materials (MXene, BFRP); periodic verification for stable materials (geopolymer aggregate, aerogel)

Tier 2: Component Testing (TEST-03 through TEST-07)

Scope: Individual products and sub-assemblies — Aegis-C panel, Phantom actuator, Charlemagne hull section, protonic transceiver, bridge bearing pad

Standards: Product-specific (UL, FCC, FMVSS, ABS, ASTM) per ROAD-05

Test Facilities: TEM cell (EMI), vibration table (LDS V8 or equivalent), environmental chamber (Thermotron or equivalent), EMC chamber (3 m semi-anechoic), high-voltage lab

Sample Rate: First article inspection (100% dimensional/functional), production sampling per AQL (Acceptable Quality Level) tables

Tier 3: System Testing (TEST-03 through TEST-07)

Scope: Integrated systems — complete Aegis Home, hardened substation, Charlemagne vessel in dry dock, Phantom MK-1 full robot, protonic network (5 nodes)
Test Conditions: Simulated Carrington environment (combined EM + thermal + mechanical per CSMTTest protocol), accelerated aging, fault injection, degraded-mode operation
Sample Rate: Every pilot unit (100%); production units sampled at 1 per 50 for EMI, 100% for functional/safety

Tier 4: Field Testing (TEST-09)

Scope: Deployed products in real operational environments — 50-home Aegis Home pilot, 10 Phantom utility deployment, Charlemagne vessel in commercial service, 5-node protonic network, infrastructure pilot sites
Data Collection: Continuous sensor monitoring per ROAD-06 pilot data collection plans
Duration: Minimum 12 months continuous data collection per pilot program
Success Criteria: Per-pilot threshold values defined in ROAD-06

STANDARD TEST FACILITIES REQUIRED

CSM-Owned Facilities (El Segundo)

Facility	Capability	Equipment	Status	Capacity
TEM Cell	EMI shielding effectiveness, 1 kHz-10 GHz, E-field and H-field probes	Custom TEM cell, Keysight N5227A PNA network analyzer, NIST-calibrated field probes	OPERATIONAL	1 test setup; capacity: 8 tests/month
Materials Test Lab	Tensile, compression, flexural, hardness, density	Instron 5982 universal test machine (100 kN), hardness testers, analytical balance, density kit	OPERATIONAL	20 specimens/day
Environmental Chamber	Temperature -70°C to +180°C, humidity 10-98% RH, programmable cycling	Thermotron SE-600-10-10 walk-in chamber	PROCUREMENT (Q4 2026)	5 m³ internal volume
Vibration Table	Sine, random, shock; 5 Hz-3,000 Hz, 50 kN force	LDS V875-440 shaker with slip table	PROCUREMENT (Q4 2026)	1 m² table area
High-Voltage Lab	DC to 100 kV, AC to 50 kV, impulse to 200 kV	Hipotronics HV test set, impulse generator	PROCUREMENT (Q1 2027)	15 m² cleared test area
XPS/XRD/SEM Suite	Materials characterization — surface chemistry, crystal structure, morphology	Thermo Scientific K-Alpha XPS, Bruker D8 Advance XRD, JEOL SEM	SHARED (University partner)	In-house procurement deferred to Phase II

Third-Party Test Facilities (Contracted)

Facility	Location	Capability	Standards	Typical Cost	Lead Time
UL Solutions	Northbrook, IL	Electrical safety, fire, weathering	UL 94, 723, 746C, 263, 1973, etc.	\$12K-\$55K per test	3-6 months
TÜV SÜD America	San Diego, CA	EMC, FCC Part 15, IEC 61000-4-x	FCC, IEC, CE marking	\$8K-\$25K	2-4 months
NTS (National		MIL-STD-461,	MIL-STD-188-125,		

Technical Systems)	Fullerton, CA	MIL-STD-810, HEMP, vibration, thermal	MIL-STD-461, MIL-STD-810	\$15K-\$180K	4-8 months
Element Materials Technology	Multiple US/EU	ASTM, ISO, materials characterization, fire	ASTM D149, D257, D3039, E84, etc.	\$3K-\$18K	1-3 months
White Sands HEMP Facility	White Sands, NM	High-Altitude EMP (HEMP) testing	MIL-STD-188-125	\$180K+	6-12 months (DoD scheduling)
ABS / DNV Surveyors	Global	Maritime classification	SOLAS, class rules	\$15K-\$35K per survey	1-3 months per survey

DATA MANAGEMENT PLAN

Data Acquisition

Measurement Type	Instrumentation	Sampling Rate	Data Format	Storage
EMI Shielding Effectiveness (SE)	Keysight PNA + field probes	1,001 points per sweep, 1 kHz-10 GHz	.s2p (Touchstone S-parameter)	CSM Digital Twin
GIC Current/Voltage	NI DAQ + shunt resistors + voltage dividers	1 Hz continuous	.tdms (NI binary) or .csv	Local NAS → cloud sync
Temperature / Humidity	K-type thermocouples, capacitive RH sensors	1 sample/minute	.csv	Cloud
Vibration (Acceleration)	PCB Piezotronics accelerometers, NI DAQ	5 kHz continuous	.tdms	Local NAS
Mechanical (Strain, Load, Displacement)	Strain gauges, LVDT, load cells, NI DAQ	100 Hz continuous	.tdms	Local NAS
Visual / Thermal Imaging	FLIR thermal camera, 4K visible camera	30 fps video	.mp4 (H.265)	NAS (video)

Data Storage Policy

- Raw data retention: Minimum 10 years (irreplaceable Carrington-equivalent test data)
- Processed data: Indefinite retention (published datasets, certification evidence)
- Backup: 3-2-1 rule — 3 copies, 2 different media, 1 off-site (AWS S3 Glacier Deep Archive for long-term)
- Encryption: AES-256 at rest, TLS 1.3 in transit
- Estimated annual data volume: Phase I (2026–2027): 50 TB; Phase II (2028–2030): 500 TB; Phase III (2031+): 2 PB

Data Access

Access Level	Who	What	Conditions
Internal — Full	CSM engineers, Zirconia, Steele	All raw and processed data	Standard employee confidentiality agreement
Internal — Redacted	CSM sales, marketing, investor relations	Processed data, summary statistics, charts	No raw data containing proprietary manufacturing details
Scientific Advisory Board	University partners	Full raw data (excluding proprietary manufacturing parameters)	Data Use Agreement; co-authorship on publications
Insurance Partners	Swiss Re, AXA actuaries	Anonymized pilot data; aggregate statistics; claims correlation	NDA + Data Use Agreement; no PII from homeowners
Public	Researchers, general	Published datasets (de-identified, aggregated), test Open Data Commons	

	public	methodology documents	Attribution License
Regulators	UL, FCC, ICC-ES, ABS surveyors	Full test data for products under evaluation	Standard regulatory access; NDA where required

REPORTING STANDARDS

Test Report Template — Required Sections

Every CSM test report shall include:

- Test Identification:** Unique test ID (e.g., CSMTest-2026-001-Aegis-C-Panel-SE), date, test engineer(s), witness(es)
- Test Objective:** Clear statement of what is being tested and why
- Device Under Test (DUT):** Description, serial number, configuration, photographs, BOM reference
- Test Standards Referenced:** All applicable standards with revision dates (e.g., IEEE 299-2006, MIL-STD-188-125-1)
- Test Equipment:** Make, model, serial number, calibration due date for all instruments
- Test Setup:** Diagram, photograph, description of test fixture, cabling, grounding
- Test Procedure:** Step-by-step procedure, including equipment settings, dwell times, number of repetitions
- Environmental Conditions:** Temperature, relative humidity, barometric pressure at time of test
- Measurement Uncertainty:** Uncertainty budget per ISO/IEC 17025, expressed as expanded uncertainty (k=2, 95% confidence)
- Results:** Raw data (appended), processed results (tables, charts), statistical analysis
- Pass/Fail Determination:** Comparison of results against pre-defined criteria with margin
- Deviations:** Any deviations from referenced standard or test plan, with justification
- Conclusions:** Summary of findings, recommendations for design changes or further testing
- Appendices:** Raw data files, calibration certificates, equipment lists, photographs

Measurement Uncertainty Budget Template

For each measurement type, CSM maintains an uncertainty budget:

Component	Source	Distribution	Standard Uncertainty (u)	Divisor	Sensitivity Coefficient (c)	u×c Contribution
Field Probe Calibration	NIST calibration certificate	Normal	U/2 (from cert)	2	1	(U/2)×1
Network Analyzer Amplitude	Keysight spec (±0.1 dB)	Rectangular	0.1/√3	√3	1	0.058 dB
Cable Loss	Measured, S21 of cables	Normal	0.05 dB	1	1	0.05 dB
Mismatch	VSWR of probe + DUT	U-shaped	Calculated	√2	1	Calculated
Repeatability	10 repeat measurements	Normal	Standard deviation	1	1	σ
Combined Standard Uncertainty (uc)			$\sqrt{(\sum(ui \times ci)^2)}$			
Expanded Uncertainty (U, k=2)			2 × uc			

Target: Expanded uncertainty for SE measurement <±3 dB at 95% confidence. Currently achieved: ±2.1 dB.

TEST PRIORITIZATION MATRIX

Given limited TEM cell capacity (8 tests/month) and budget constraints, test prioritization follows:

Priority	Criterion	Example Tests
P0 — CRITICAL	Required for regulatory approval (UL, FCC, ICC-ES), safety certification, or blocking a TRL gate	Aegis Home UL 94/723/746C; Phantom Phase 1 EM; FCC Part 15 for LoRa; ICC-ES AC04
P1 — HIGH	Required for pilot program initiation, Series A/B funding gate, or customer commitment	Aegis Embark TEM cell; Charlemagne hull SE; Protonic H-FET; Aegis Home prototype full-system GIC test
P2 — MEDIUM	Performance optimization, competitive benchmarking, publication data	Stellar-Tint mode comparison; Way-Finder haptic accuracy; Silence-Block cabin noise optimization
P3 — LOW	Nice-to-have data for marketing, internal R&D, exploratory	Long-term MXene degradation (5-year outdoor); Phantom actuator wear beyond 10 ⁶ cycles; YInMn Blue colorfastness (10-year equivalent)

TEST EXECUTION GOVERNANCE

Roles and Responsibilities

Role	Individual	Responsibilities
Test Director	Zirconia	Approves all test plans; assigns test priority; signs off on test reports; ensures NIST traceability and ISO 17025 compliance
Test Engineer	Mork (lead), rotating by product	Designs test setup; executes test; collects data; performs initial analysis; drafts test report
Quality Assurance	Chester	Witnesses critical tests; reviews test reports for completeness; manages non-conformance process; maintains calibration schedule
Product Owner (Agent)	Per ROAD-07 mapping	Defines test objectives and pass/fail criteria; provides DUT and product expertise; reviews and approves test reports for their product
Independent Reviewer	Scientific Advisory Board member (rotating)	Reviews test methodology for P0/P1 tests; may witness testing; co-authors publications

Test Plan Approval Workflow

Test Engineer drafts plan →
Product Agent reviews (technical accuracy) →
Zirconia reviews (methodology, standards compliance) →
Chester reviews (quality, calibration) →
Independent Reviewer reviews (for P0/P1) →
Zirconia approves →
Test executed →
Test Engineer writes report →
Chester reviews report →
Product Agent approves →
Zirconia signs off →
Report published to CSM Digital Twin

NON-CONFORMANCE MANAGEMENT

When test results fail to meet acceptance criteria:

- 1. **Immediate Actions:** DUT quarantined; test equipment verified (re-calibrate if needed); test repeated to rule out measurement error
- 2. **Root Cause Analysis:** 5-Why analysis or Ishikawa (fishbone) diagram; trace failure to material, design, manufacturing, or test setup root cause
- 3. **Corrective Action:** Design change (ECO), manufacturing process change (Work Instruction update), supplier corrective action request (SCAR), or test procedure revision
- 4. **Verification:** Corrected DUT re-tested; if pass, corrective action validated; if fail, return to Step 2
- 5. **Documentation:** Non-Conformance Report (NCR) filed in CSM QMS; linked to test report; trended monthly for systemic issues
- 6. **Communication:** For P0/P1 tests, non-conformance reported to Director Steele within 24 hours; mitigation plan within 72 hours

CONTINUOUS IMPROVEMENT — TEST PROCESS METRICS

Metric	Current (Q3 2026)	Target (Q4 2027)	Improvement Action
Test report cycle time (plan to signed report)	8 weeks (estimated)	4 weeks	Standardized templates; pre-approved test setups for routine tests; parallel review workflow
Measurement uncertainty (SE)	±2.1 dB (k=2)	±1.5 dB (k=2)	Improved cable calibration; better anechoic treatment of TEM cell; automated uncertainty calculator
First-pass yield (tests passing on first attempt)	N/A (insufficient data)	>85%	Better pre-test simulation (CST model predicts SE before physical test); design reviews catch issues before test
Calibration overdue rate	0% (all current)	<2% (with scaling)	Automated calibration management software; redundant instruments to allow calibration without downtime
Independent review completion	0% (not yet established)	100% for P0/P1	Scientific Advisory Board engagement (2027); rotating reviewer schedule

TEST FACILITY BUDGET — 5-YEAR PROJECTION

Facility	2026 (Q3-Q4)	2027	2028	2029	2030	5-Year Total
CSM El Segundo Lab (equipment)	\$850K (chamber, shaker)	\$450K (HV lab, XPS)	\$200K (upgrades)	\$150K	\$100K	\$1.75M
CSM El Segundo Lab (operations)	\$180K	\$350K	\$500K	\$600K	\$750K	\$2.38M
Third-Party Testing (UL, TÜV, NTS, etc.)	\$500K	\$1.2M	\$2.5M	\$3.5M	\$5.0M	\$12.7M
Calibration Program	\$35K	\$65K	\$100K	\$150K	\$200K	\$550K
Data Management	\$45K	\$120K	\$250K	\$400K	\$600K	\$1.42M

(storage, software)						
Scientific Advisory Board	\$0	\$105K	\$105K	\$105K	\$105K	\$420K
ANNUAL TOTAL	\$1.61M	\$2.29M	\$3.66M	\$4.91M	\$6.76M	\$19.22M

“Testing is the conscience of engineering. A design can be elegant on paper, cost-optimized in a spreadsheet, and beautiful in a rendering — and still fail in the TEM cell. The Master Test Plan exists to ensure that every CSM product that reaches a customer has been tested to destruction in the laboratory so that it will never be destroyed in the field. NIST traceability means our numbers are real. Peer review means our claims are credible. Open data means our results are verifiable. These are not optional. They are the price of admission to the Dielectric Citadel.” — Zirconia, Director of Test & Validation

Document generated 2026-07-15. This Master Test Plan is the governing document for all CSM testing activities. All product-specific test protocols (TEST-02 through TEST-10) derive their authority from this plan and must be consistent with its requirements. Revision controlled via CSM QMS document control system. Next review: 2026-10-01 (Q4 2026), or upon any significant change to test facilities, standards, or regulatory requirements.